

Descriptive Statistics

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Warning

This work is currently under progress.

This page presents the preliminary analysis done on the dataset.

Loading the data

First, we want to load the different datasets :

```
library(foreign)
hbm = suppressWarnings(
  read.spss("data_management/data/raw/wave_1/KEN_2022_K-LSRH_v01_M_SPSS/hbm.sav",
    to.data.frame = TRUE))
hr = suppressWarnings(
  read.spss("data_management/data/raw/wave_1/KEN_2022_K-LSRH_v01_M_SPSS/hr.sav",
    to.data.frame = TRUE))
rr = suppressWarnings(
  read.spss("data_management/data/raw/wave_1/KEN_2022_K-LSRH_v01_M_SPSS/rr.sav",
```

```

    to.data.frame = TRUE))
wer = suppressWarnings(
  read.spss("data_management/data/raw/wave_1/KEN_2022_K-LSRH_v01_M_SPSS/wer.sav",
    to.data.frame = TRUE))
dim(rr)

```

```
[1] 9285  939
```

```
dim(hr)
```

```
[1] 9304  955
```

```
dim(hhm)
```

```
[1] 45011  147
```

```
dim(wer)
```

```
[1] 7244  218
```

From the [documentation](#), we have that :

Data file	Cases	Variables
hhm This file contains data at the level of the individual for all household members	45011	147
hr This file contains household level information	9304	955
rr This file contains data at the representative respondent level	9285	939
wer This file contains data for the women empowerment module	7244	218

Which seems to confirm the good loading of the data.

Age distribution

If we interest ourselves in the first data file `hbm` with information at the individual level of all surveyed household members, we can obtain the distribution of individuals across age.

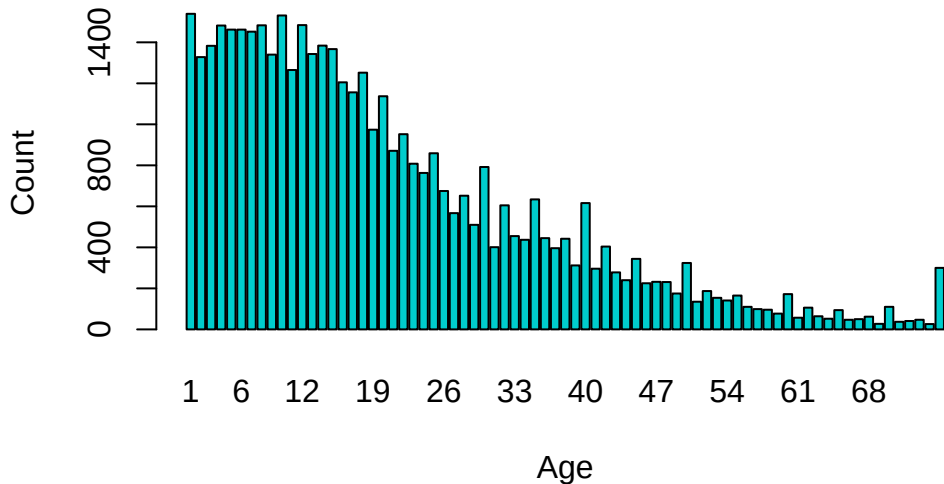
First, if we want the total distribution of individuals across age :

```
maincol = "cyan3"
table(hbm$a4a_ageyrs)
```

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1539	1328	1383	1482	1462	1462	1452	1483	1340	1531	1265	1484	1343	1384	1367	1205
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1156	1252	974	1137	871	952	808	763	859	675	567	652	510	792	401	605
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
455	437	634	445	396	442	312	616	296	404	278	240	344	225	232	231
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64
175	324	135	187	154	141	165	110	99	96	77	172	57	106	64	52
65	66	67	68	69	70	71	72	73	74	75+					
94	47	50	62	27	110	37	41	47	26	300					

```
plot(hbm$a4a_ageyrs,
     xlab = "Age", ylab = "Count",
     main = "Total distribution across age",
     col = maincol)
```

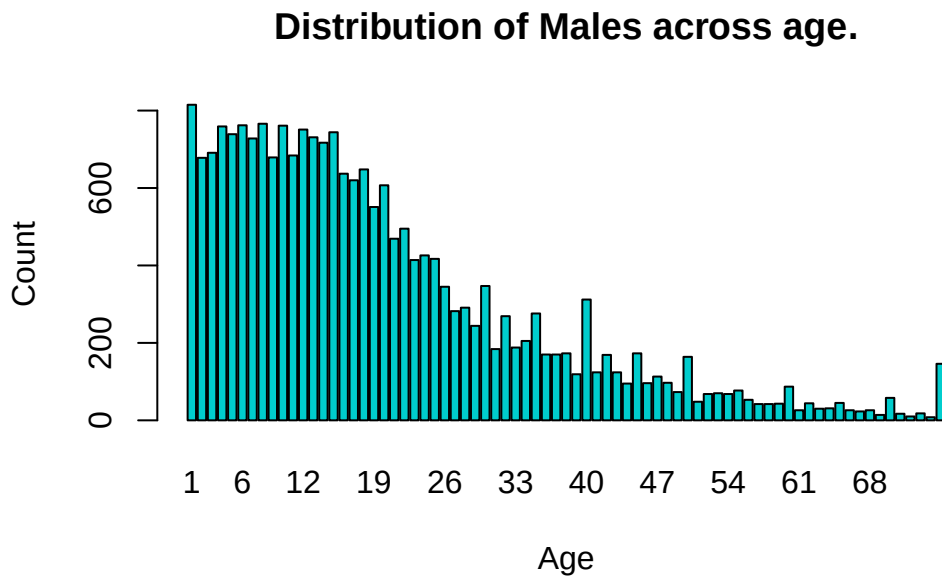
Total distribution across age



Note that the last plotted value of the `hbm$a4a_ageyrs` variable is the NA value. This behavior is reproduced in all following plots of this variable.

If we want to plot genders separately. For males, we can do :

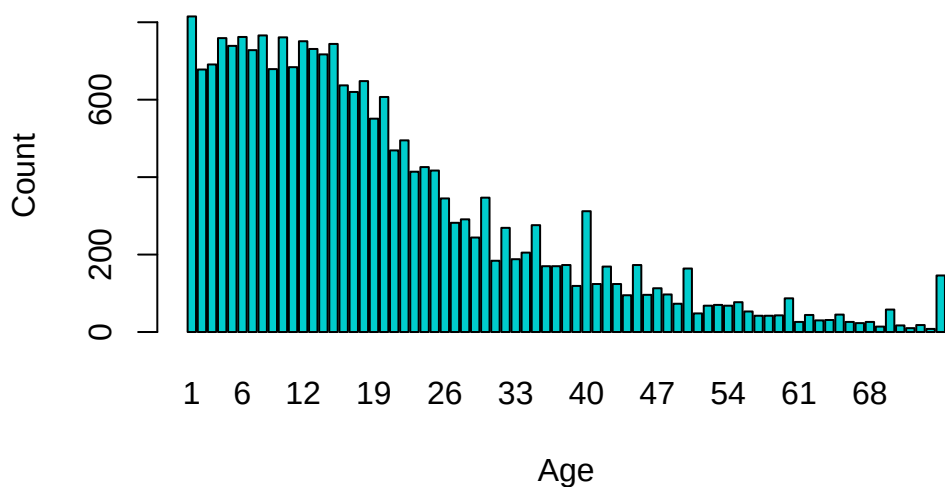
```
hbm_male = hbm[hbm$a4a_gender == "Male",]  
plot(hbm_male$a4a_ageyrs,  
     xlab = "Age", ylab = "Count",  
     main = "Distribution of Males across age.",  
     col = maincol)
```



For females, we have :

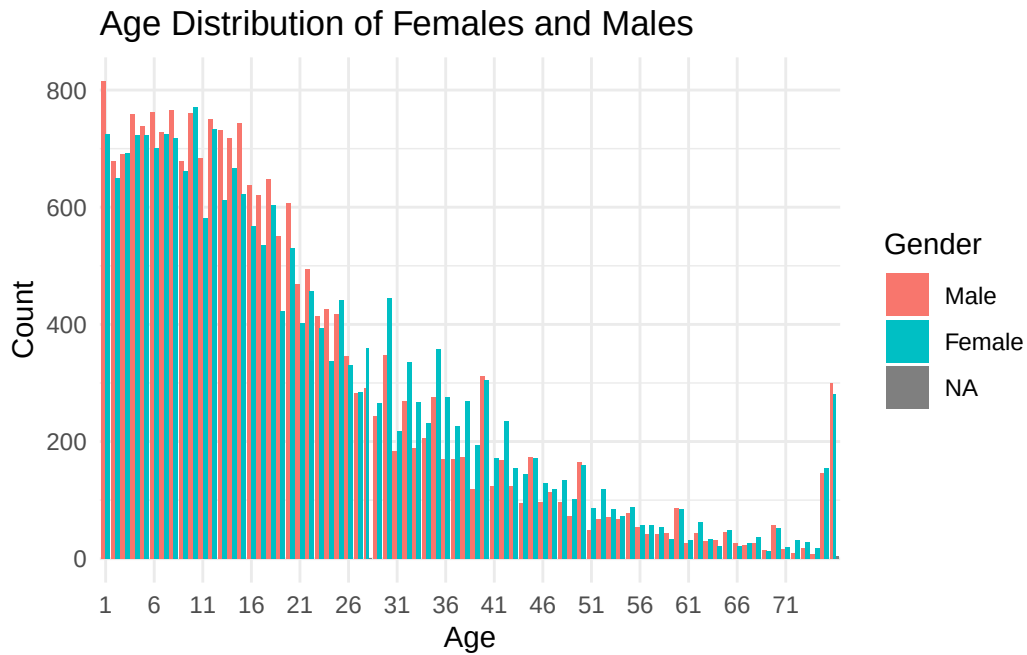
```
hbm_female = hbm[hbm$a4a_gender == "Female",]  
plot(hbm_female$a4a_ageyrs,  
     xlab = "Age", ylab = "Count",  
     main = "Distribution of Females across age.",  
     col = maincol)
```

Distribution of Females across age.



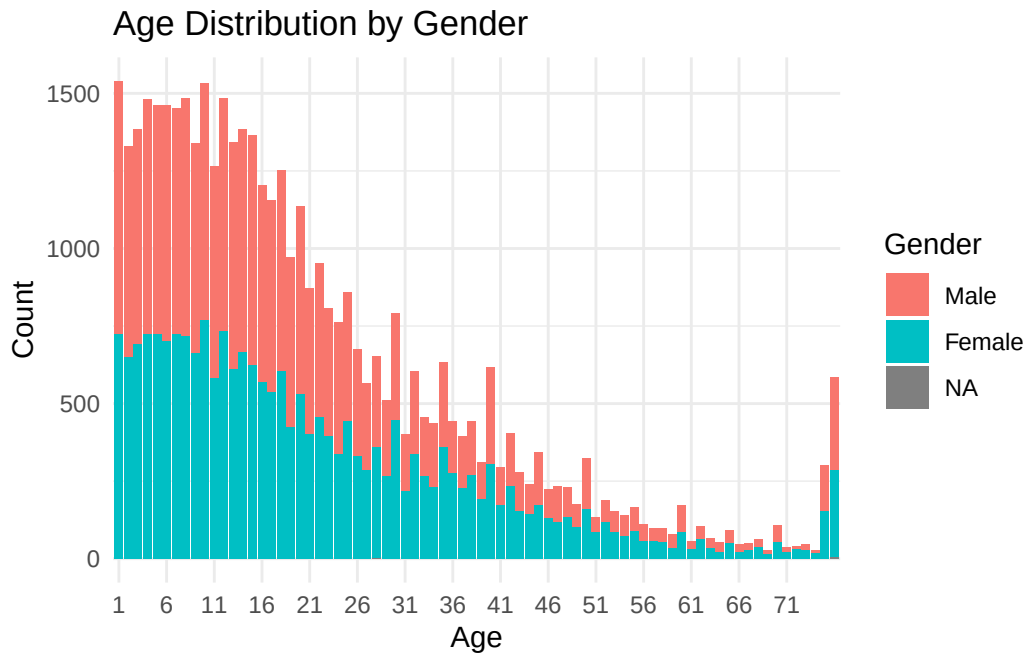
Plotted together, we have :

```
library(ggplot2)
ggplot(hhm, aes(x = a4a_ageyrs, fill = a4a_gender)) +
  geom_bar(position = "dodge") +
  theme_minimal() +
  xlab("Age") +
  ylab("Count") +
  scale_fill_discrete(name = "Gender") +
  ggtitle("Age Distribution of Females and Males") +
  scale_x_discrete(breaks = seq(min(as.numeric(hhm$a4a_ageyrs), na.rm = TRUE),
    max(as.numeric(hhm$a4a_ageyrs), na.rm = TRUE), by = 5))
```



Or alternatively :

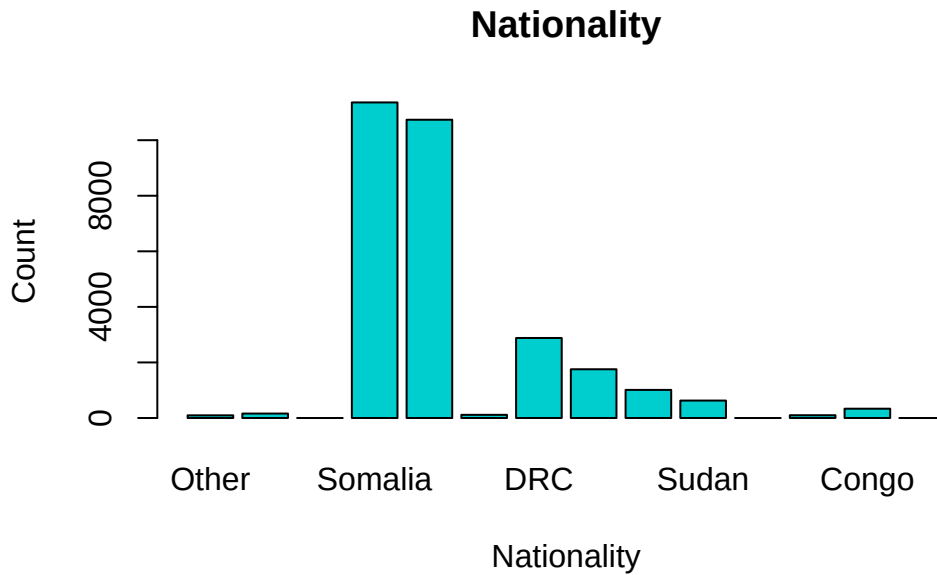
```
ggplot(hhm, aes(x = a4a_ageyrs, fill = a4a_gender)) +
  stat_count() +
  theme_minimal() +
  xlab("Age") +
  ylab("Count") +
  scale_fill_discrete(name = "Gender") +
  ggtitle("Age Distribution by Gender") +
  scale_x_discrete(breaks = seq(min(as.numeric(hhm$a4a_ageyrs), na.rm = TRUE),
    max(as.numeric(hhm$a4a_ageyrs), na.rm = TRUE), by = 5))
```



Nationality Distribution

There are 12 different nationalities identified among the surveyed individuals. Represented with bars, we get :

```
plot(hhm$a4a_nationality,  
     xlab = "Nationality", ylab = "Count",  
     main = "Nationality",  
     col = maincol)
```



To get an interactive pie chart, we could do :

```

{r}
library(plotly)
plot_ly(data = hhm,
        labels = levels(hhm$a4a_nationality),
        values = table(hhm$a4a_nationality),
        type = 'pie')

```

⚠ Warning

Plotly does not render in pdf, but is available on the [website](#). I choose not to include a standard static pie chart due to small values making the format less viable for visualisation.

Migration status distribution

There are 8 possible values for migration status in the survey :

```
levels(hhm$a4a_migration)
```

```

[1] "Don't Know"
[2] "Declined"

```

```

[3] "Other"
[4] "Citizen"
[5] "Resident"
[6] "Refugee"
[7] "Asylum seeker (does not yet hold refugee status)"
[8] "Migrant with right to work"

```

However, it seems that three of them have 0 individuals registered :

```
table(hhm$a4a_migration)
```

```

              Don't Know
              0
            Declined
              0
              Other
              5
            Citizen
            14097
            Resident
            1708
            Refugee
            28561
Asylum seeker (does not yet hold refugee status)
              616
      Migrant with right to work
              0

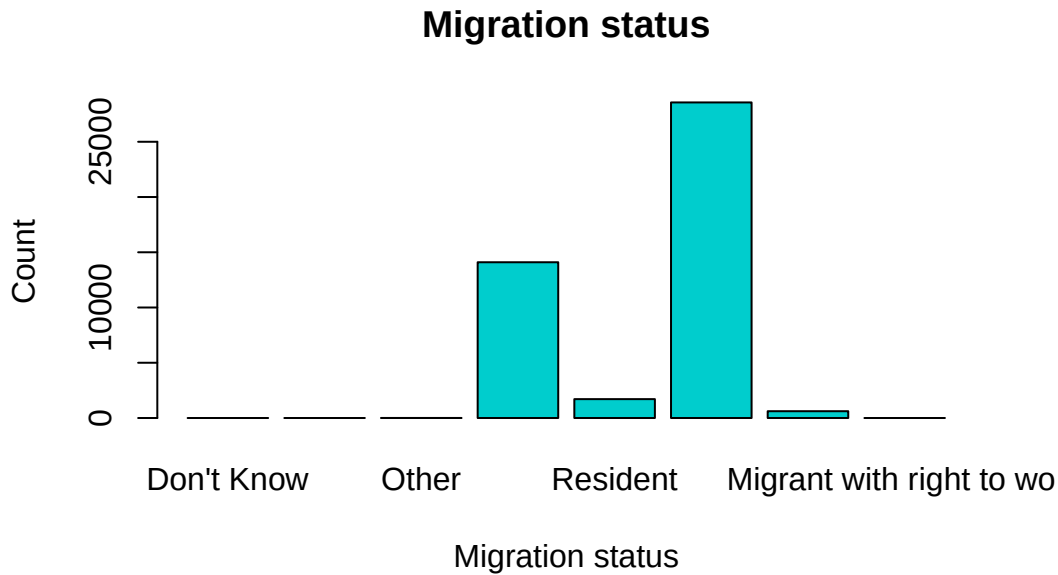
```

Representing with bars, we get :

```

# Making the name shorter so that it is included in the plot :
levels(hhm$a4a_migration)[7] = "Asyleum seeker"
plot(hhm$a4a_migration,
     xlab = "Migration status", ylab = "Count",
     main = "Migration status",
     col = maincol)

```



To do an interactive pie plot, we could do :

```

{r}
plot_ly(data = hhm,
  labels = levels(hhm$a4a_migration),
  values = table(hhm$a4a_migration),
  type = 'pie')

```

Warning

Plotly does not render in pdf, but is available on the website. I choose not to include a standard static pie chart due to small values making the format less viable for visualisation.

To come

The further analysis is still planned :

- Joint distribution of the mentioned variables.

We are interested in the gap of consumption between Refugees and Resident. We are therefore going to include other consumption variables in the following analysis.